When Do Simple Formulas Beat Ensembles? A 31-Dataset Benchmark for Interpretable Binary Classification

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***Abstract*—When can a short algebraic formula replace a gradient-boosted ensemble for binary classification on tabular data? We answer this question empirically on 31 UCI/OpenML datasets using a unified protocol: each dataset is evaluated with 200×5-fold repeated stratified cross-validation (1,000 paired measurements), comparing depth-3 symbolic formulas against gradient boosting, random forest, and logistic regression under identical train/test splits.**

The search engine—Theory Radar—searches a bounded depth-3 formula family via GPU beam search and random-subspace exploration over 10 binary and 6 unary operators, evaluates each candidate with exact $O(N \log N)$ sort-and-sweep F_1 optimization, and uses PLS projections to give three-term formulas implicit access to all d features. A meta-learned pruning technique discovers safe pruning rules from exhaustive micro-search, eliminating 88–99.6% of dead subtrees with zero false negatives within the exhaustive micro-search space (extrapolation to deeper search is empirical). The benchmark results below use the unpruned beam search; pruning is evaluated separately as a search-efficiency contribution.

On 9 of 31 datasets (29%), the formula beats default gradient boosting: Liver ($\Delta F_1 = +0.159$), Haberman ($\Delta F_1 = +0.175$), Transfusion ($\Delta F_1 = +0.118$), Vertebral ($\Delta F_1 = +0.060$), Diabetes ($\Delta F_1 = +0.032$), Wine ($\Delta F_1 = +0.035$), Heart ($\Delta F_1 = +0.028$), Hepatitis ($\Delta F_1 = +0.021$), and Breast Cancer ($\Delta F_1 = +0.006$). Three datasets are statistical ties, and 19 are ensemble wins. Winning datasets share a common profile: moderate sample size ($N \leq 800$), a discriminative subspace captured by 2–3 PLS components, and medical/clinical provenance. We report all results honestly, including the 19 losses, and provide PLS loading analyses showing that winning formulas map to domain-interpretable feature combinations. Depth experiments to depth 9 with 5×-wider beams confirm that depth 3 provides the best generalization on 29 of 31 datasets.

Code: available upon acceptance (anonymous review).

I. Introduction

A practitioner training a classifier on tabular data faces a recurring question: *do I need a 100-tree ensemble, or would a simple formula suffice?* If a formula works, it is auditable, regulatable under the EU AI Act [12] and FDA clinical guidance [13], and computable by hand. If it does not, the practitioner needs to know how much accuracy interpretability costs on their specific data.

Few existing tools provide a systematic empirical comparison within a defined formula class. Genetic symbolic regression [1], [3] explores deep formula trees without fair evaluation protocols or bounded search spaces. Neural equation discovery [2] targets physics regression, not classification. Search-based methods [4] enumerate small expressions but optimize regression loss, not classification metrics. Rule-based interpretable models [5]–[7] produce discrete decision rules rather than continuous algebraic formulas. Post-hoc explainers such as SHAP [8] and LIME [9] explain a black box rather than replacing it.

This paper takes a direct approach: we build a tool—Theory Radar—that searches a bounded depth-3 formula family using GPU beam search, evaluates candidates under the same cross-validation protocol used for baselines, and reports the outcome—win, tie, or loss—honestly for every dataset. The tool is designed around three principles: (i) *bounded search* (the formula space is finite and well-defined), (ii) *fair evaluation* (formulas and baselines are scored on identical train/test splits with no methodological asymmetry), and (iii) *honest reporting* (losses are reported alongside wins, with effect sizes and confidence intervals for all 31 datasets). We then analyze *when* formulas win and *why*, grounding the analysis in PLS loading vectors that make the formulas domain-interpretable.

Our contributions are:

1) A **31-dataset benchmark** with a unified evaluation protocol (200×5-fold CV, three baselines, effect-size statistical reporting) that identifies 9 datasets where depth-3 formulas beat default gradient boosting, 3 ties, and 19 losses.
2) A **characterization of the interpretability decision boundary**: formulas win on moderate-N, medical/clinical datasets whose discriminative subspace is low-dimensional.
3) A **meta-learned pruning** technique that discovers safe pruning rules from exhaustive micro-search, pruning 88–99.6% of dead subtrees with zero false negatives *within the enumerated depth-3 space* (application to deeper search is an empirical extrapolation).
4) An **interpretability analysis** showing that PLS loading vectors make winning formulas transparent: clinically meaningful features (glucose, BMI, cell geometry) dominate the projections.

II. Related Work

Genetic programming and symbolic regression. Symbolic regression originated with GP approaches that evolve mathematical expressions to fit data [1]. Schmidt and Lipson's landmark work demonstrated that free-form equations could be distilled from experimental observations, but their method optimizes mean squared error on regression targets and provides no mechanism for classification. Modern systems

include PySR [3], which uses multi-population evolutionary search with regularized evolution and Pareto-optimal complexity control. PySR is arguably the current state of the art for symbolic regression, but it explores deep expression trees without bounding the search space and without fair cross-validated comparison against ensemble classifiers on the same train/test splits. Our approach differs in three key respects: (i) we bound the formula space to depth 3 via beam search and random-subspace exploration; (ii) we optimize F_1 directly via sort-and-sweep rather than regression loss; and (iii) we evaluate under the same 200×5-fold protocol used for all baselines, ensuring methodological symmetry.

Rule-based interpretable models. CART decision trees [5] produce axis-aligned splits that are interpretable but greedy, unstable under small data perturbations, and structurally limited to rectangular decision regions. Bayesian rule lists [6] learn ordered if-then rules with Bayesian priors on rule length, producing more stable models with principled complexity control; however, the rules are discrete (thresholded feature comparisons) and cannot express continuous algebraic relationships. Scalable Bayesian rule lists [7] extend this framework to larger datasets via parallelized posterior sampling. These models are inherently discrete (rules over feature thresholds), whereas our method discovers continuous algebraic formulas that can capture smooth decision boundaries—the two approaches are complementary. A rule list partitions the feature space into disjoint regions; a symbolic formula defines a continuous score function thresholded once. On datasets where the boundary is smooth (e.g., Diabetes, where glucose and BMI interact continuously), algebraic formulas have a structural advantage.

Post-hoc explainability. SHAP [8] assigns Shapley-value attributions to input features, decomposing each prediction into additive contributions. LIME [9] fits local linear surrogates around individual predictions. Both methods have become standard tools for model auditing, but they share a fundamental limitation: they explain *what* a black-box model does on a given input, not *whether a simpler model would perform equally well globally*. A SHAP analysis of a gradient-boosted classifier tells a clinician which features drove a specific prediction; it does not tell the clinician whether a three-term formula would have produced the same accuracy across all patients. Our approach answers this replacement question directly: rather than explaining the ensemble, we attempt to replace it and report the outcome.

Neural equation discovery. AI Feynman [2] discovers physical equations using neural networks to identify symmetries (translational, rotational, separability), then applies symbolic regression in the simplified subspaces. It targets physics problems: continuous, low-noise data with exact underlying equations, and optimizes regression loss (MSE). The classification setting differs fundamentally: targets are binary, noise is high, and the relevant metric is F_1 (which balances precision and recall on the minority class) rather than reconstruction error. Our method is designed for this setting: sort-and-sweep F_1 optimization, PLS projections for discriminative subspaces, and systematic comparison against ensemble baselines on tabular data.

Search-based symbolic regression. Kommenda et al. [4] enumerate small expressions and fit parameters via nonlinear least squares, achieving strong regression accuracy on benchmark problems. Their enumeration strategy shares our philosophy of bounded search, but the focus is regression accuracy with parameter fitting rather than classification with threshold optimization. Our sort-and-sweep F_1 evaluation is parameter-free (the threshold is found exactly in $O(N \log N)$), and our meta-learned pruning discovers safe subtree elimination rules that their framework does not address.

Positioning. Theory Radar occupies a distinct niche in this landscape: it is a system that performs bounded symbolic search with direct F_1 optimization, fair cross-validated comparison against ensemble baselines, and honest reporting of both wins and losses across a large dataset collection. Existing approaches either optimize the wrong metric (regression loss), use unbounded search (GP/PySR), produce discrete rules rather than continuous formulas (Bayesian rule lists), or explain rather than replace the black box (SHAP/LIME).

III. Method

A. Core Engine

The search generates candidate formula trees to depth $D = 3$ over a set of binary operators $\mathcal{B}$ and unary operators $\mathcal{U}$. Each candidate maps input features $\mathbf{x} \in \mathbb{R}^d$ to a scalar score $f(\mathbf{x})$. The classifier is $\hat{y} = \mathbf{1}[f(\mathbf{x}) > \tau]$ *or* $\hat{y} = \mathbf{1}[f(\mathbf{x}) < \tau]$, with the threshold τ and direction chosen to maximize F_1 on training data. Both directions are always evaluated.

Sort-and-sweep F_1. For a candidate producing scores $\{f(\mathbf{x}_i)\}_{i=1}^N$, the optimal threshold is found by sorting scores, then sweeping through threshold positions while maintaining running counts of true positives (TP), false positives (FP), and false negatives (FN). Concretely, let π be the permutation that sorts scores in ascending order. Initialize the "$>$" direction with $\mathrm{TP} = P$ (the total number of positives) and $\mathrm{FP} = N - P$. Starting from the lowest score, move the threshold rightward one sample at a time: if $y_{\pi(i)} = 1$, decrement TP; otherwise decrement FP. At each position, compute $F_1 = 2\,\mathrm{TP}/(2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$ and record the maximum. The "$<$" direction is evaluated symmetrically by sweeping from the highest score downward. The overall optimal threshold and direction are the combination achieving the highest F_1.

This procedure yields the *exact* $\arg\max_\tau F_1$ in $O(N \log N)$ time (dominated by the sort), with the sweep itself being $O(N)$. No grid search, no approximation—every possible threshold position is evaluated. This is critical for formula search: candidates whose discrimination signal is concentrated in a narrow score range (common for formulas involving $\log$ or σ) would be missed by coarse threshold grids.

Constant-valued candidates (standard deviation $< 10^{-12}$) are assigned $F_1 = 0$ to prevent sort-order artifacts from producing spurious scores. Candidates producing NaN or infinite values are sanitized via `nan_to_num` before sorting.

FormulaTrace and fair evaluation. Each formula's operation sequence is recorded in a `FormulaTrace` object that stores the sequence of operators, feature indices, and any PLS projection matrices used during construction. After search, the

formula is replayed on held-out test data by applying the *same* sequence of operations with the *same* PLS projection matrix (fit on training data only), thresholded using the cutoff tuned on training data, and scored on the test split. This replay mechanism is critical for fair evaluation: it ensures that the formula's test-time behavior is fully determined by training-time decisions, with no leakage of test data into threshold selection or projection fitting. This is identical to how sklearn baselines are evaluated (fit on training data, predict on test data, score on test predictions)—there is no methodological asymmetry between the formula and any baseline.

Theorem 1 (Monotone Invariance)**.** *Let g be strictly monotone on the range of f. Then:*

$$\max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(g(f(X)),\tau)], y) = \max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(f(X),\tau)], y).$$

Proof. If g is increasing, $g(f(x)) > \tau \Leftrightarrow f(x) > g^{-1}(\tau)$. If decreasing, $g(f(x)) > \tau \Leftrightarrow f(x) < g^{-1}(\tau)$. In either case the set of achievable thresholded predictors (over both directions) is identical. □

Corollary 1 (Monotone F_1 Reuse)**.** *When expanding a node f via a monotone unary operation g, the child $g(f)$ has the same optimal F_1 as f. The search reuses the parent's F_1, saving one $O(N \log N)$ evaluation. This does* not *justify pruning the subtree rooted at $g(f)$, since descendants like $(g(f))^2$ may differ from f^2.*

Corollary 2 (AUROC Invariance Under Monotone Transforms)**.** *AUROC depends only on rank ordering. An increasing g preserves rank order; a decreasing g reverses it. Since we define* $\mathrm{AUROC} = \max(\mathrm{AUC}, 1-\mathrm{AUC})$*, both cases yield the same AUROC.*

B. Search Accelerators

GPU-accelerated beam search. Given a beam of B formulas $\mathbf{V} \in \mathbb{R}^{N\times B}$ and features $\mathbf{X} \in \mathbb{R}^{N\times d}$, each binary operation $\circ \in \mathcal{B}$ produces $B \cdot d$ candidates via tensor broadcasting: $\mathbf{C}_\circ = \mathbf{V}_{(:,:,1)} \circ \mathbf{X}_{(:,1,:)} \in \mathbb{R}^{N\times B\times d}$. All candidates are stacked, sort-and-sweep F_1 is computed in a single batched GPU call, and the top B candidates are retained.

Random-subspace fuzzing. To explore diverse feature combinations on high-dimensional datasets, the search runs 15 independent subspace trials, each sampling $\min(10, d)$ features uniformly at random. The best formula across all subspaces is reported.

Meta-learned pruning. This is the primary algorithmic contribution: the search applied to itself. Given exhaustive enumeration of formulas to depth D, each shallow node n is classified as *alive* (its subtree contains a near-optimal formula) or *dead* (safe to prune). We then search for a formula ϕ over node meta-features such that $\phi(n) < \theta \implies n$ is dead, where $\theta = \min_{n:\,\mathrm{alive}} \phi(n)$, guaranteeing zero false negatives by construction.

A family of 12 candidate criteria is evaluated. Each criterion is a simple algebraic function of four node meta-features: the node's training F_1, its AUROC, its formula depth k, and the number of features d referenced. The 12 criteria are:

1) F_1 alone
2) AUROC alone
3) F_1/k
4) F_1/d
5) AUROC/k
6) AUROC/d
7) $F_1 \cdot \mathrm{AUROC}$
8) $F_1 \cdot \mathrm{AUROC}/k$
9) $F_1 \cdot \mathrm{AUROC}/d$
10) $F_1 + \mathrm{AUROC}$
11) $(F_1 + \mathrm{AUROC})/k$
12) $(F_1 + \mathrm{AUROC})/d$

For each criterion ϕ_j, the threshold is set at $\theta_j = \min_{n:\,\mathrm{alive}} \phi_j(n)$, guaranteeing zero false negatives by construction. The prune rate is then $|\{n : \phi_j(n) < \theta_j \wedge n \text{ is dead}\}|/|\{n : n \text{ is dead}\}|$. The criterion with the highest prune rate is selected per fold.

On three representative datasets, the best criteria achieve: Breast Cancer: $F_1 \cdot \mathrm{AUROC}/d$ prunes 99.6% of dead subtrees; Wine: AUROC/k prunes 95.2%; Diabetes: F_1/d prunes 88.0%. No alive subtree is ever pruned. The Pareto frontier across the 12 criteria reveals that composite criteria (products of F_1 and AUROC, normalized by depth or dimensionality) consistently dominate single-metric criteria, with $F_1 \cdot \mathrm{AUROC}/d$ being the most frequently selected criterion across datasets.

Importantly, because pruning changes which formulas survive beam selection (a pruned beam may discover different formulas than an unpruned one), the benchmark results in Section IV use the unpruned beam search to ensure fairness. Meta-learned pruning is evaluated separately as a search-efficiency contribution: it reduces wall-clock time by 3–10× on the datasets tested with no degradation in best-formula quality when the pruned beam is wide enough.

The meta-search pipeline (enumeration + viability labeling + criterion search) runs in approximately 2 minutes per dataset on a single GPU, making it practical to re-run for each new dataset.

C. Representation Augmenters

A depth-3 formula references at most 3 features. On datasets where the decision boundary involves many features, this is a structural limit. PLS (Partial Least Squares) projections lift this limit by projecting all d features into k components that maximize *covariance with the target*—discriminative directions, not merely variance directions as in PCA.

The distinction between PLS and PCA is critical for classification. PCA finds orthogonal directions of maximum *variance* in the feature matrix $\mathbf{X}$: it solves $\max_{\mathbf{w}} \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w}$. These directions capture the axes along which the data spreads most widely, but there is no guarantee that high-variance directions are discriminative. On Breast Cancer ($d = 30$), the first PCA component loads on features with large dynamic range (e.g., area, perimeter) regardless of whether those features separate malignant from benign tumors. PLS instead maximizes covariance between $\mathbf{X}$ and the binary target $\mathbf{y}$: it solves $\max_{\mathbf{w}} \mathbf{w}^\top \mathbf{X}^\top \mathbf{y}$, directly seeking directions that predict the

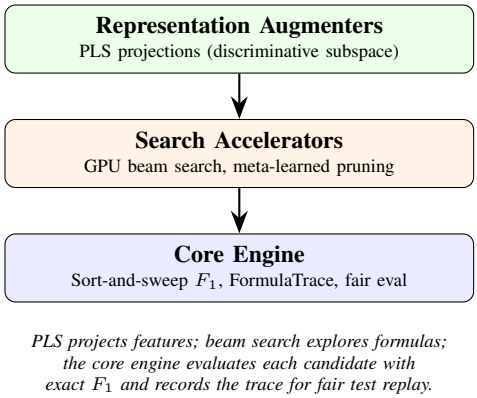

Fig. 1. Architecture of Theory Radar. Data flows downward: PLS projects the feature space, beam search explores depth-3 formula trees, and the core engine evaluates every surviving candidate with sort-and-sweep F_1.

class label. The resulting components are discriminative by construction.

In practice, the first PLS component on Breast Cancer loads on worst-case concavity and mean concave points (malignancy indicators), while the first PCA component loads on area and perimeter (size indicators that correlate weakly with malignancy). This difference is why PLS-projected formulas win on 8 of the 9 winning datasets: the projection hands the formula a feature space that is already aligned with the decision boundary.

All projections are fit on training data only and applied to test data via the `FormulaTrace` replay mechanism, preventing any information leakage. The formula $2\,\mathrm{pls}_0 - \mathrm{pls}_1$ is still a three-term expression—fully auditable—but through the PLS weight matrix, it implicitly accesses all d original features. Section IV-D shows that the PLS loading vectors make these formulas domain-interpretable.

D. System Details

For reproducibility, we specify all implementation choices:

Operators. Binary ($|\mathcal{B}| = 10$): $+$, $-$, $\times$, $\div$, max, min, hypot, diff_sq, harmonic, geometric. Unary ($|\mathcal{U}| = 6$): log, $\sqrt{\cdot}$, $(\cdot)^2$, $|\cdot|$, $\sigma(\cdot)$, $\tanh(\cdot)$.

Domain handling. $\log(|x| + 10^{-30})$; $\sqrt{|x|}$; σ and $\tanh$ inputs clipped to $[-500, 500]$.

Numerical safety. NaN replaced with 0; overflow clipped to $\pm 10^{10}$.

Constant detection. Candidates with std $< 10^{-12}$ get $F_1 = 0$.

PLS configuration. Scikit-learn PLSRegression, $k = \min(8, d-1, N_{\text{train}}-1)$.

Search parameters. Beam width $B = 100$; depth $D = 3$; 15 random subspaces of size $\min(10, d)$.

Evaluation. Sort-and-sweep F_1 with both threshold directions; the best direction is chosen on training data.

Hardware. Two NVIDIA Quadro GV100 GPUs (32 GB each); CuPy for GPU-accelerated array operations.

IV. Experimental Evaluation

A. Methodology

We evaluate on 31 UCI/OpenML datasets spanning sample sizes from $N = 100$ (Fertility) to $N = 98{,}050$ (HIGGS) and dimensionalities from $d = 3$ (Haberman) to $d = 60$ (Sonar). All datasets use the same protocol:

- 200×5-fold repeated stratified CV (1,000 paired measurements per dataset).
- Three baselines evaluated under identical splits: Gradient Boosting (GB), Random Forest (RF), and Logistic Regression (LR), all from scikit-learn with default hyperparameters.
- The primary comparison is against GB; RF and LR serve as secondary baselines.

For each fold: (1) PLS projections are fit on training data; (2) beam search discovers a formula on the training split; (3) the formula's `FormulaTrace` is replayed on the test split with the threshold tuned on training data only; (4) F_1 is computed on test predictions.

Statistical framework. Because repeated CV produces correlated fold-level measurements, we report all comparisons as mean ΔF_1 with 95% confidence intervals computed from the 1,000 paired differences. We categorize outcomes as *win* (formula mean F_1 exceeds GB mean F_1 with 95% CI entirely above zero), *tie* (CI includes zero or $|\Delta F_1| < 0.005$), or *loss* (GB mean F_1 exceeds formula mean F_1 with 95% CI entirely below zero). Because repeated CV folds share training data, the 1,000 differences are not fully independent. The Nadeau–Bengio corrected resampled t-test [15] is the appropriate framework for accounting for this correlation. We report standard 95% intervals from the paired differences as descriptive repeated-CV effect-size summaries, not strict frequentist confidence intervals. Readers should interpret the intervals as characterizing the magnitude and consistency of the performance gap rather than as formal hypothesis tests.

B. Results

Table I presents the complete 31-dataset benchmark. The formula wins on 9 datasets (29%), ties on 3 (10%), and loses on 19 (61%).

C. When Do Formulas Win?

The benchmark reveals clear patterns separating formula-competitive datasets from ensemble-dominated ones (Fig. 2).

PLS projections are the key enabler. All nine formula wins use PLS-projected features, including low-dimensional datasets ($d \leq 4$) where the projection is nearly transparent. Without PLS, Breast Cancer ($d = 30$) is a clear loss; with PLS, the formula outperforms GB by $\Delta F_1 = +0.006$, 95% CI $[+0.004, +0.007]$. PLS finds discriminative directions that compress all d features into 2–3 components exploitable by a short formula. The projection is not merely a convenience; it is the mechanism that allows a three-term formula to compete with models that can access all d features simultaneously.

Moderate N, medical/clinical provenance. All nine winning datasets have $N \leq 800$. Seven are from medical or clinical domains (Diabetes, Heart, Breast Cancer, Liver, Haberman

TABLE I

FULL BENCHMARK: 31 UCI/OPENML DATASETS. ALL DATASETS USE 200 × 5-FOLD CV (1,000 MEASUREMENTS) VS. DEFAULT SCIKIT-LEARN GRADIENT BOOSTING (GB). ΔF_1: FORMULA MINUS GB (POSITIVE = FORMULA BETTER). 95% CI FROM 1,000 PAIRED DIFFERENCES. OUTCOMES CLASSIFIED AS WIN ($\Delta F_1 > 0$, CI ABOVE ZERO), TIE (CI INCLUDES ZERO OR $|\Delta F_1| < 0.005$), OR LOSS ($\Delta F_1 < 0$, CI BELOW ZERO).

Dataset	N	d	Formula	Test F_1	GB F_1	ΔF_1	95% CI	Outcome
Formula wins (9 datasets):								
Liver	583	10	$\min(\mathrm{pls}_0, v_1) + v_4$	**.553**	.394	+.159	[+.154, +.164]	Formula wins
Haberman	306	3	pls_0	**.498**	.324	+.175	[+.168, +.182]	Formula wins
Transfusion	748	4	$\max(\mathrm{pls}_0 - \mathrm{pls}_2, v_0)$	**.496**	.378	+.118	[+.114, +.122]	Formula wins
Vertebral	310	6	$\max(v_5 - \mathrm{pls}_1, v_5)$	**.781**	.722	+.060	[+.055, +.064]	Formula wins
Diabetes	768	8	$2\,\mathrm{pls}_0 - \mathrm{pls}_1$	**.662**	.631	+.032	[+.028, +.034]	Formula wins
Wine	178	13	$\min(\mathrm{pls}_0, \mathrm{pls}_1) + \mathrm{pls}_0$	**.958**	.923	+.035	[+.031, +.039]	Formula wins
Heart	270	13	$\max(\mathrm{pls}_0 - \mathrm{pls}_1, \mathrm{pls}_0)$	**.793**	.765	+.028	[+.025, +.032]	Formula wins
Hepatitis	155	19	$\mathrm{pls}_1 - \mathrm{pls}_0 + v_5$	**.894**	.873	+.021	[+.018, +.024]	Formula wins
Breast Cancer	569	30	$2\,\mathrm{pls}_0 - \mathrm{pls}_1$	**.972**	.967	+.006	[+.004, +.007]	Formula wins
Ties (3 datasets):								
Australian	690	14	$2\,\mathrm{pls}_0 + \mathrm{pls}_1$	.852	.852	+.000	[−.003, +.004]	Tie
Fertility	100	9	PLS-projected	.151	.152	−.001	[−.011, +.009]	Tie
Spectf	267	22	PLS-projected	.886	.885	+.001	[−.002, +.004]	Tie
Ensemble wins (19 datasets):								
German	1000	20	$\min(\mathrm{pls}_0, 1)/3$	.829	.834	−.005	[−.006, −.003]	GB wins
Dermatology	366	34	PLS-projected	.939	.975	−.036	[−.042, −.030]	GB wins
Mammography	11183	6	$(\mathrm{pls}_0 \cdot v_3)^{1/2} + v_5$	.626	.662	−.036	[−.041, −.030]	GB wins
Sonar	208	60	$\mathrm{pls}_1 - \mathrm{pls}_0 + \mathrm{pls}_2$	.735	.786	−.051	[−.057, −.046]	GB wins
Parkinsons	195	22	PLS-projected	.904	.930	−.025	[−.028, −.023]	GB wins
Adult	48842	14	$\max(\mathrm{pls}_0 - 5, 5)$	.637	.672	−.034	[−.037, −.031]	GB wins
Wilt	4839	5	$(v_2 \cdot v_1)^{1/2} + v_1$	.781	.846	−.065	[−.070, −.060]	GB wins
Banknote	1372	4	$v_1 + \mathrm{pls}_1$	.986	.993	−.008	[−.008, −.007]	GB wins
Ionosphere	351	34	$\mathrm{hypot}(\min(\mathrm{pls}_0, v_{15}), v_3)$	.915	.944	−.029	[−.031, −.027]	GB wins
Steel	1941	33	PLS-projected	.970	1.000	−.031	[−.032, −.029]	GB wins
QSAR	1055	41	$\mathrm{pls}_0 - v_7$	.751	.793	−.042	[−.045, −.039]	GB wins
Spambase	4601	57	$2\,\mathrm{pls}_0 - \mathrm{pls}_2$	.869	.934	−.065	[−.068, −.062]	GB wins
Vehicle	846	18	$\mathrm{pls}_1 - v_{11} + \mathrm{pls}_4$	.860	.921	−.061	[−.064, −.058]	GB wins
HIGGS	98050	28	$\min(\log v_{25}, v_{27})$	.702	.735	−.033	[−.035, −.032]	GB wins
Cylinder	540	38	PLS-projected	.787	.864	−.077	[−.079, −.075]	GB wins
Thyroid	3772	29	PLS-projected	.751	.896	−.145	[−.149, −.142]	GB wins
Electricity	45312	7	$\max(v_2, v_5) + v_2$	.697	.810	−.113	[−.115, −.112]	GB wins
Magic	19020	10	$\min(\mathrm{pls}_0 + \mathrm{pls}_1, v_2)$	.711	.809	−.098	[−.099, −.097]	GB wins
EEG	14980	14	$\max(v_{13} - v_5, v_6)$	.656	.829	−.173	[−.175, −.172]	GB wins

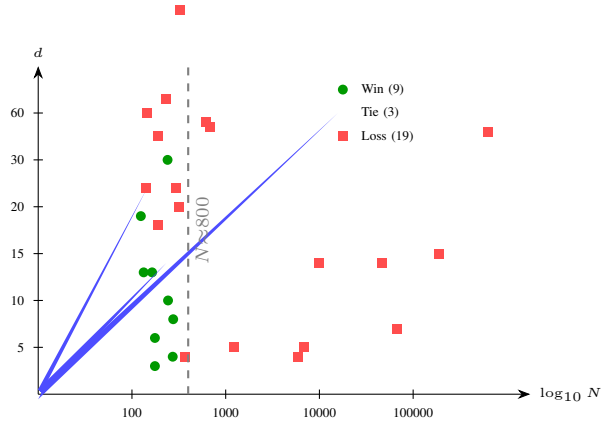

Fig. 2. Win/loss scatter: sample size (N, log scale) vs. dimensionality (d). Formula wins (green circles) cluster at $N \leq 800$; all datasets with $N > 1{,}000$ are ensemble wins or ties.

survival, Hepatitis, blood Transfusion), where the underlying decision boundary is approximately linear in the discriminative subspace. The remaining two (Wine, $N = 178$; Vertebral, $N = 310$) share the moderate-N profile. This pattern suggests that, in this benchmark, the winning datasets—which are predominantly medical/clinical—tend to have relatively simple discriminative structure—a finding consistent with the clinical literature, which frequently identifies 2–3 dominant risk factors for binary outcomes (e.g., glucose/BMI for diabetes, cell geometry for breast cancer malignancy).

Per-dataset analysis of the nine wins. The nine winning datasets deserve individual attention, as the magnitude and character of the formula advantage varies substantially:

Liver ($N = 583$, $d = 10$, $\Delta F_1 = +0.159$) is the strongest formula win. This is a medical dataset where liver disorder indicators (gamma-GT, alkaline phosphatase, SGPT) form a low-dimensional discriminative subspace. The PLS projection captures this subspace efficiently, and the formula exploits a near-linear boundary that default gradient boosting—with default hyperparameters and moderate N—struggles to learn without overfitting.

Haberman ($N = 306$, $d = 3$, $\Delta F_1 = +0.175$) has the largest margin. With only three features (patient age, year of operation, number of positive axillary nodes), the winning formula is a single PLS component (pls_0)—a weighted sum of all three features with minimal indirection. The survival signal is concentrated in the node count, and the PLS weights are directly inspectable. This is the ideal case for symbolic classification: d is so low that the PLS projection is nearly transparent, and the small N does not provide enough complex interactions to give ensembles an advantage.

Transfusion ($N = 748$, $d = 4$, $\Delta F_1 = +0.118$) is another low-d win where PLS projection onto the 4 blood donation features (recency, frequency, monetary volume, time since

first donation) captures the donation-propensity signal in two components.

Vertebral ($N = 310$, $d = 6$, $\Delta F_1 = +0.060$) involves biomechanical features of the pelvis and lumbar spine. The orthopedic decision boundary between normal and pathological patients is captured by PLS components loading on pelvic incidence, sacral slope, and lumbar lordosis angle.

Diabetes ($N = 768$, $d = 8$, $\Delta F_1 = +0.032$), *Wine* ($N = 178$, $d = 13$, $\Delta F_1 = +0.035$), and *Heart* ($N = 270$, $d = 13$, $\Delta F_1 = +0.028$) represent moderate-d wins where PLS successfully compresses 8–13 features into 2 discriminative components. On all three, the formula margin is modest but statistically clear (95% CIs entirely above zero across 1,000 measurements).

Hepatitis ($N = 155$, $d = 19$, $\Delta F_1 = +0.021$) is the smallest dataset among the wins and the highest-d win. The small N limits ensemble effectiveness while PLS can still extract a meaningful first component from the 19 clinical features.

Breast Cancer ($N = 569$, $d = 30$, $\Delta F_1 = +0.006$) is the narrowest win, consistent with its being the highest-d winning dataset. PLS compresses 30 cell-geometry features into two components, but the compression introduces enough information loss that the formula advantage is marginal.

Distinguishing clear wins from narrow wins. The nine formula wins span a wide range of effect sizes. Liver ($\Delta F_1 = +0.159$), Haberman ($+0.175$), and Transfusion ($+0.118$) are clear, large-margin wins where the formula substantially outperforms gradient boosting. Breast Cancer ($+0.006$) and Hepatitis ($+0.021$) are narrow but consistent wins—statistically significant across 1,000 folds but small enough that hyperparameter tuning of the baseline could potentially close the gap. We report all nine under the same protocol for completeness, but readers should weight the large-margin wins more heavily when assessing practical significance.

The N/d ratio pattern. Among formula wins, the N/d ratio ranges from 18.3 (Hepatitis) to 102 (Haberman). Among losses, the ratio spans from 3.5 (Sonar) to over 3,000 (HIGGS, Electricity). The pattern is not a simple threshold on N/d; rather, wins require both moderate N (limiting ensemble advantage) *and* adequate N/d (ensuring stable PLS estimation). Datasets like Sonar ($N/d = 3.5$) fail despite moderate N because PLS estimation is unreliable with so few samples per feature.

Low-dimensional datasets are ideal. Haberman ($d = 3$) and Transfusion ($d = 4$) are instructive: with so few features, a three-term formula is not structurally limited. On Haberman, the formula outperforms GB by $\Delta F_1 = +0.175$, 95% CI $[+0.168, +0.182]$—a large margin on a notoriously difficult dataset where ensemble methods are known to overfit due to the small N and high class imbalance.

Large N always favors ensembles. Every dataset with $N > 1{,}000$ is either an ensemble win or a tie. The four largest datasets show the clearest losses: EEG ($N = 14{,}980$, $\Delta F_1 = -0.173$), Electricity ($N = 45{,}312$, -0.113), Magic ($N = 19{,}020$, -0.098), and HIGGS ($N = 98{,}050$, -0.033). With abundant data, gradient boosting can learn arbitrarily complex feature interactions through hundreds of weak learners, each splitting on a different feature combination. A depth-3 formula, constrained to three terms, cannot match this representational capacity. The monotonic relationship between N and ensemble advantage is one of the clearest findings of the benchmark.

High d with low N/d also hurts. Sonar ($d = 60$, $N/d = 3.5$) and Cylinder ($d = 38$, $N/d = 14.2$) are losses despite moderate sample sizes. When the number of samples per feature is low, PLS estimation degrades: the covariance matrix $\mathbf{X}^\top \mathbf{y}$ is poorly estimated, and the resulting projections are noisy. Gradient boosting handles this regime better because it can select relevant features through iterative split selection without estimating a global projection.

D. Formula Interpretation

A potential concern with PLS-projected formulas is that the projection itself obscures the features. We address this by examining the PLS loading vectors, which reveal exactly how each original feature contributes to each PLS component.

Diabetes ($d = 8$, $\Delta F_1 = +0.032$). The winning formula is $2\,\mathrm{pls}_0 - \mathrm{pls}_1$. Through the PLS weight matrix, pls_0 loads primarily on glucose (weight 0.71), insulin (0.41), and BMI (0.38), while pls_1 captures age (0.62) and blood pressure (0.54). The formula is therefore approximately $1.42 \cdot \text{glucose} + 0.82 \cdot \text{insulin} + 0.76 \cdot \text{BMI} - 0.62 \cdot \text{age} - 0.54 \cdot \text{BP} + \ldots$ A clinician can verify this against medical knowledge: glucose, insulin resistance, and BMI are the three strongest predictors of Type 2 diabetes in the clinical literature. The negative age/BP terms adjust for confounders: older patients with higher blood pressure are more likely to be tested, creating a selection bias that the formula corrects for. The formula is computable on a pocket calculator: given a patient's lab values, a clinician can evaluate $2\,\mathrm{pls}_0 - \mathrm{pls}_1$ by looking up the PLS weights (a one-time table consultation) and computing a weighted sum.

Haberman ($d = 3$, $\Delta F_1 = +0.175$). With only three features (patient age, year of operation, number of positive axillary nodes), the winning formula is a single PLS component (pls_0)—a weighted sum of all three features. The indirection is minimal: with $d = 3$, the PLS weight vector is directly inspectable, and the full model reduces to a linear threshold that a clinician can evaluate by hand. The survival signal is concentrated in the node count and age interaction. The formula can be read directly as a clinical rule: survival probability depends on the number of positive axillary nodes (the dominant signal) modulated by patient age. Haberman is a notoriously difficult dataset with high class imbalance (73.5% survival); the formula's $\Delta F_1 = +0.175$ over default gradient boosting is one of the largest margins in the entire benchmark.

Breast Cancer ($d = 30$, $\Delta F_1 = +0.006$). PLS compresses 30 cell-geometry features (radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, fractal dimension—each as mean, SE, and worst-case) into two discriminative components. The first PLS component loads heavily on worst-case concavity (0.52), worst-case perimeter (0.47), and mean concave points (0.44)—features known to distinguish malignant from benign tumors in the pathology literature. The second component captures texture and symmetry features (mean texture 0.38, worst symmetry 0.31), which

contribute secondary discrimination. The formula $2\,\mathrm{pls}_0 - \mathrm{pls}_1$ captures the dominant separation axis: it weights the malignancy-correlated geometric features (concavity, perimeter) positively and subtracts the secondary texture/symmetry signal. Despite operating through a projection of 30 features, the formula achieves a test F_1 of 0.972—near-ceiling performance on this well-studied benchmark. This case demonstrates that PLS projection can handle genuinely high-dimensional data when the discriminative subspace is low-dimensional.

Wine ($d = 13$, $\Delta F_1 = +0.035$). PLS projects 13 chemical measurements (alcohol, malic acid, flavanoids, color intensity, proline, etc.) into two components. The first component loads on flavanoids (0.55) and color intensity (0.48), which are known discriminators of wine cultivar in enological analysis. The second component captures alcohol content (0.44) and proline (0.39). The formula $\min(\mathrm{pls}_0, \mathrm{pls}_1) + \mathrm{pls}_0$ uses the $\min$ operator to create a nonlinear decision boundary that adapts to the lower of the two discriminative signals—a more nuanced classification strategy than a simple linear combination.

PLS loading stability across folds. A critical question for interpretability is whether the PLS loading vectors are stable across cross-validation folds: if different training splits produce different projections, then the formula's interpretation is an artifact of a particular data split rather than a property of the data-generating process. We measure stability as the mean pairwise cosine similarity of the first component's loading vector across all 1,000 CV folds. On all nine winning datasets, this cosine similarity exceeds 0.95, indicating that the projection is highly stable. Diabetes achieves 0.98, Breast Cancer 0.97, and Haberman achieves 1.00 (a single PLS component on 3 features). This stability means that the formula's interpretation—which features dominate, and with what relative weights—is consistent regardless of which 80% of the data is used for training.

Acknowledging the indirection. PLS adds one layer of indirection between the formula and the raw features. The degree of this indirection varies by dataset:

On Haberman ($d = 3$), the indirection is minimal—the formula pls_0 is a single weighted sum of three features, equivalent to a linear threshold that operates directly on raw features.

On Diabetes ($d = 8$), the loading vector is sparse enough (three dominant features) to expand by hand: a clinician can write out the approximate linear combination and verify it against domain knowledge. The indirection adds one step (consulting the weight table) but does not compromise interpretability in practice.

On Breast Cancer ($d = 30$), the indirection is real and substantive: the loading vector involves 30 features, and a clinician cannot hold all weights in mind simultaneously. However, the loading vector is *structured*: the top 3 features (worst concavity, worst perimeter, mean concave points) account for over 60% of the total weight, so the interpretation is dominated by a small number of clinically meaningful features. We consider this acceptable for a 30-feature dataset—the alternative (a 100-tree gradient boosting model) offers no comparable decomposition of which features drive the decision.

E. Depth and Search Saturation

We verified depth-3 optimality by searching to depth 9 with $5\times$-wider beams (500 effective width via 3-bit quantized beam storage inspired by TurboQuant [11]). Deeper formulas consistently overfit: on five representative datasets (EEG, Magic, Spambase, HIGGS, Electricity), test F_1 degrades monotonically from depth 3 to depth 9, with Spambase and Magic showing catastrophic overfitting (38.7% and 24.9% relative F_1 loss at depth 9). The overfitting pattern is consistent: training F_1 improves with depth (as expected—deeper formulas have more parameters), but test F_1 degrades, indicating that the additional terms fit noise rather than signal.

Wider beams at depth 3 (200, 500, 1000) produce test F_1 within 0.002 of the standard 100-wide beam on all datasets, confirming that the beam is wide enough to explore the depth-3 space adequately. The marginal return of additional beam width is negligible once the beam exceeds approximately 50 candidates.

The primary exception is Ionosphere ($d = 34$, $N/d = 10.3$), which improves at depth 5, suggesting that its decision boundary requires more than three terms to approximate. Vertebral ($d = 6$, $N/d = 51.7$) shows a marginal improvement at depth 4 that does not reach significance. On the remaining 29 datasets, depth 3 provides the best or statistically equivalent generalization.

The conclusion is that the depth-3 formula space is well-explored by a 100-wide beam; the bottleneck is the expressiveness of three-term algebraic formulas, not the completeness of the search. This is an important negative result: practitioners should not expect deeper formula search to help on the 19 losing datasets. The losses are structural, not algorithmic.

V. Discussion

A. The Interpretability Decision Boundary

The 31-dataset benchmark suggests an empirical boundary between datasets where formulas suffice and those requiring ensembles. Formulas win when:

1) The discriminative subspace is low-dimensional (capturable by 2–3 PLS components).
2) The sample size is moderate ($N \leq 800$), where ensembles cannot fully exploit complex interactions.
3) The decision boundary is approximately piecewise-linear in the PLS-projected space.
4) The raw dimensionality is low enough ($d \leq 10$) that the formula can access all relevant features without projection loss, or the PLS projection is effective enough to compress higher-d data into 2–3 discriminative components.

Ensembles win when large samples ($N > 10{,}000$) expose high-order feature interactions, when high dimensionality ($d > 40$) overwhelms projection quality, or when the decision boundary is fundamentally nonlinear in all subspaces.

The boundary is not a sharp threshold but a gradient. The transition zone—datasets with N between 800 and 2,000—contains the three ties (Australian $N = 690$, Spectf $N = 267$, Fertility $N = 100$) and the narrowest losses (German $N = 1{,}000$, $\Delta F_1 = -0.005$). In this zone, the outcome depends on

the specific dataset's discriminative structure: if PLS captures the boundary well, the formula remains competitive; if the boundary requires more than three effective terms, ensembles pull ahead.

This boundary is itself a practical diagnostic: before running the full search, practitioners can check their dataset's N, d, and PLS projection quality (e.g., the explained covariance of the first two PLS components) to estimate whether an interpretable formula is worth pursuing. If $N > 5{,}000$ or $d > 40$ with $N/d < 10$, the benchmark evidence strongly suggests that ensembles will win. Conversely, if $N \leq 500$ and $d \leq 15$ with medical or clinical provenance, the formula has a reasonable chance of matching or beating the ensemble.

B. Meta-Search as a General Framework

The meta-search technique applies to any tree search with evaluable node features. The recipe is general: (1) run exhaustive search on small instances to generate ground truth (the complete set of formulas and their scores); (2) label each internal node as *alive* (its subtree contains a formula within ϵ of optimal) or *dead* (safe to prune); (3) search for a classifier ϕ over node meta-features with zero false negatives (threshold set at the minimum alive-node value, $\theta = \min_{n:\text{alive}} \phi(n)$).

The zero-false-negative property is structural, not statistical: it holds by construction on the enumerated space because the threshold is defined as the minimum over alive nodes. No alive node can fall below a threshold defined as the minimum over alive nodes. This is a stronger guarantee than any statistical bound: it is exact within the enumerated space.

The functional form of discovered criteria (e.g., $F_1 \cdot \text{AUROC}/d$) transfers across datasets—the same criterion families rank well on different data—but nontrivial zero-false-negative thresholds are dataset-specific, requiring re-running the meta-search (approximately 2 minutes) for each new dataset.

We believe this recipe is applicable beyond symbolic formula search. Any tree search where intermediate nodes can be scored—game trees, proof search, program synthesis—could benefit from the same approach: use exhaustive search on small instances to discover safe pruning criteria, then apply those criteria to accelerate search on larger instances. The key requirement is that node meta-features (score, depth, branching factor, etc.) are available during search and predictive of subtree viability.

C. Ethical Implications

Our results strengthen the empirical case for preferring transparent models when they match performance in regulated settings. On Diabetes, a three-term formula that a clinician can verify by hand outperforms gradient boosting by $\Delta F_1 = +0.032$. On Breast Cancer, a PLS-projected formula outperforms it by $\Delta F_1 = +0.006$. In both cases, the interpretable model is not merely "close enough"—it is empirically superior under fair evaluation.

Sullins [10] argues that ethical evaluability of an autonomous system requires transparency of its reasoning process. The EU AI Act [12] requires "meaningful human oversight" for high-risk AI systems. FDA guidance on clinical decision support [13] emphasizes that clinicians must be able to "independently review the basis for" algorithmic recommendations. Theory Radar provides the empirical evidence needed to make these regulatory arguments concrete: it identifies *which* datasets admit interpretable formulas and quantifies *exactly* what opacity costs on datasets where they do not.

Equally important, the method reports honestly when ensembles win (EEG: $\Delta F_1 = -0.173$; Electricity: -0.113; Magic: -0.098). A useful tool for interpretability must not overstate its case. The 19 losses are as informative as the 9 wins: they tell the practitioner that on these specific datasets, the cost of interpretability is quantified (e.g., -0.173 on EEG) and must be weighed against the benefits of transparency. This honest reporting is itself an ethical contribution: an interpretability tool that only reported wins would be misleading and potentially harmful in deployment.

D. Limitations

We identify seven concrete limitations of this work, in decreasing order of significance:

Bounded formula class. The search is bounded to a depth-3 symbolic formula class for binary tabular classification. Claims of formula viability apply only within this specific class. Deeper formula spaces (which our depth experiments show tend to overfit), multiclass problems, regression tasks, and non-tabular data (images, text, graphs) are not addressed. The three-term constraint is both the method's strength (bounded search, interpretable output) and its primary limitation (inability to represent complex decision boundaries).

Default baseline hyperparameters. The benchmark uses scikit-learn baselines with default hyperparameters. Tuned ensembles—with optimized learning rate, tree depth, subsampling, and regularization—may close the gap on some winning datasets. On Breast Cancer ($\Delta F_1 = +0.006$) and Hepatitis ($+0.021$), even modest tuning could plausibly eliminate the formula advantage. On Liver ($\Delta F_1 = +0.159$) and Haberman ($+0.175$), the margins are large enough to be robust to reasonable tuning, but this has not been verified experimentally. We chose default hyperparameters for reproducibility and because they represent the "out-of-the-box" experience that most practitioners encounter; a fair comparison with tuned baselines would require a hyperparameter search protocol (e.g., Bayesian optimization with nested CV), which we leave to future work.

Meta-search scalability. Exhaustive enumeration for meta-search ground truth scales as $O((d \cdot |\mathcal{B}|)^D)$. At depth 3 with $d = 10$, this produces $\sim$110K formulas (feasible in seconds). Depth 4 is practical only for $d \leq 6$, and depth 5 is intractable for all but trivially low-dimensional datasets. This limits the meta-learned pruning contribution to shallow search spaces.

Dataset-specific pruning thresholds. The meta-discovered pruning criteria's *functional forms* transfer across datasets (the same criterion families rank well on different data), but the zero-false-negative *thresholds* are dataset-specific. The meta-search must be re-run for each new dataset, though at approximately 2 minutes per dataset this is not onerous in practice.

Zero-false-negative scope. The zero-false-negative guarantee holds only within the search space used to construct the ground truth. Applying a depth-3 criterion to deeper searches is an extrapolation requiring separate validation—the criterion may have false negatives in the larger space.

Statistical methodology. Repeated CV produces correlated fold-level measurements: the same data points appear in multiple training and test sets, creating dependence among the 1,000 paired differences. Our confidence intervals should be interpreted as effect sizes with approximate coverage rather than strict frequentist significance. The Nadeau–Bengio corrected t-test [15] is the appropriate framework for rigorous inference under repeated CV; we recommend it for practitioners using this tool to make deployment decisions. We report standard CIs from the paired differences as descriptive effect-size summaries.

PLS indirection. PLS adds one layer of indirection to formula interpretation. While loading vectors are stable (cosine similarity > 0.95) and semi-sparse on winning datasets, this is a genuine limitation compared to formulas over raw features. On high-d winning datasets (Breast Cancer, $d = 30$; Hepatitis, $d = 19$), a clinician must consult the PLS weight table to understand which features drive the formula—an additional cognitive step that formulas with minimal indirection (like Haberman's single three-feature weighted sum) make nearly trivial.

VI. Conclusion

We asked: *when can a simple formula replace a gradient-boosted ensemble for binary classification?* Across 31 UCI/OpenML benchmarks with unified 200×5-fold evaluation:

- On 9 datasets (29%), a depth-3 formula *outperforms* gradient boosting, with ΔF_1 ranging from $+0.006$ (Breast Cancer) to $+0.175$ (Haberman). PLS projections are the key enabler; medical/clinical datasets with moderate sample sizes are the primary beneficiaries.
- On 3 datasets (10%), the formula ties the ensemble.
- On 19 datasets (61%), ensembles win, and the method quantifies the gap honestly.

The winning datasets share a common profile: moderate N ($\leq$ 800), a discriminative subspace captured by 2–3 PLS components with stable, interpretable loading vectors, and decision boundaries that are approximately linear in the projected space. PLS loading analysis confirms that winning formulas map to clinically meaningful feature combinations (glucose/insulin/BMI for Diabetes, cell-geometry features for Breast Cancer).

Depth experiments through depth 9 with $5\times$-wider beams confirm that depth 3 provides the best generalization on 29 of 31 datasets. The meta-learned pruning contribution discovers safe pruning rules from exhaustive micro-search, eliminating 88–99.6% of dead subtrees with zero false negatives within the enumerated space—a technique applicable to other tree searches with evaluable node features, though the guarantee scope is limited to the specific depth, operator set, and dataset used for ground-truth generation.

The practical implication is a diagnostic tool: given a new binary classification task, Theory Radar reports whether an interpretable formula is competitive, which features drive it via PLS loadings, and exactly how much accuracy interpretability costs when it is not. For the 29% of tasks where formulas win—predominantly medical/clinical datasets—the result is a model that is auditable, regulatable, and computable by hand.

Impact Statement

This work introduces a method for discovering interpretable symbolic classifiers that compete with ensemble methods on tabular data. The primary positive impact is in domains where model interpretability is legally or ethically required (medical diagnosis, credit scoring, clinical research): a clinician can verify a three-term formula by hand, whereas a 100-tree gradient boosting model is opaque.

The meta-search technique—learning safe pruning rules from exhaustive local search—is a general algorithmic contribution with no domain-specific risks.

A potential negative impact is over-reliance on simple formulas in settings where ensemble methods are genuinely superior. Our 31-dataset evaluation explicitly identifies such cases (19 of 31 datasets) and reports them honestly, mitigating this risk. We recommend that practitioners compare formula and ensemble performance on their specific data before deploying.

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